

[Date: 2026-05-01]

Editor-in-Chief *npj Precision Oncology*

Dear Editor,

We submit our manuscript “**An 8-gene RAI-responsiveness biomarker outperforms BRAF V600E status in papillary thyroid carcinoma**” for consideration as an Article. This v7 version supersedes v4/v6: in addition to the previously submitted multi-cohort meta-analysis, Decision Curve Analysis, subgroup forest, and browser-based interactive calculator, v7 incorporates the results of a 10-round multi-omic audit (R1–R10) executed during the pre-submission window. The audit added: a structural-variant analysis showing 76.8% RET-fusion enrichment in DM1, arm-level CNA showing chr 7p/7q gain DM1-specificity (bootstrap 95% CI [0.000, 0.079], one-sided $p = 0.053$), TPO promoter methylation Cohen’s $d = 2.30$ with methylation $\times$ expression Spearman $\rho = -0.68$, an independent Cabrita 2020 12-gene TLS replication in two cohorts (TCGA $d = +0.67$ $p = 4.4 \times 10^{-15}$, GSE286332 $d = +1.96$ $p = 1.4 \times 10^{-4}$), and a 3-way clinical pilot algorithm in which **DM1 + RET-fusion-positive + TERT-WT (n = 67) shows 0 OS events** — a candidate biomarker-positive cohort for a continued-RAI-responder protocol. The TERT, decision-curve, meta-analytic, subgroup-forest, and GSE76039 external-validation results from v4/v6 are unchanged.

Decisions about radioactive-iodine (RAI) therapy in papillary thyroid carcinoma (PTC) currently rely on BRAF V600E mutation status and clinician-driven judgement about differentiation, with no quantitative pre-treatment biomarker that outperforms the mutation alone. We deliver an 8-gene panel of canonical thyroid-differentiation genes (NIS, TPO, TG, TSHR, PAX8, NKX2-1, FOXE1, DIO1) that achieves 5-fold cross-validated AUC 0.954 in DM1/DM2 cluster prediction (Random Forest 0.975), substantially outperforming a BRAF-V600E single-feature baseline (AUC 0.822, $\Delta\text{AUC} = +0.132$) on 513 TCGA-THCA primary tumours.

Three reviewer-anticipated robustness layers added in v4 (Fig. 6E–G). 1. **Multi-cohort meta-analysis** across four independent transcriptomic cohorts (GSE27155 $n = 99$, GSE29265 $n = 49$, GSE33630 $n = 105$, GSE76039 $n = 37$; total $n = 290$). Random-effects logit-AUC pooling yields **pooled AUC = 0.980 (95% CI 0.869–0.997, $I^2 = 0\%$, all 4 cohorts AUC > 0.96)**. The $I^2 = 0\%$ is a notably tight result for a thyroid expression biomarker — most prior transcriptomic panels show $I^2 > 50\%$. 2. **Decision Curve Analysis** (Vickers & Elkin 2006) of 8-gene LogReg vs BRAF V600E vs treat-all/treat-none across

decision thresholds 0.05–0.95 ($n = 91$). The 8-gene strategy is the **dominant strategy at all 91 thresholds**, the strongest possible Decision Curve outcome — translating the AUC outperformance into a clinical-utility statement. 3. **9-strata subgroup forest** across Age (<45, 45–65), Sex (Male, Female), Stage (I/II, III/IV), and Histology (cPTC, FVPTC) with bootstrap 95% CIs. **7 of 9 strata exceed AUC 0.85**; the highest-risk Stage III/IV subgroup peaks at **AUC 0.996** — exactly the subgroup pre-RAI decision-making is hardest in.

Browser-based interactive RAI calculator (reports/v17_boost/rai_calculator.html) — a self-contained client-side JavaScript implementation that embeds the trained LogReg coefficients and the TCGA-THCA reference distribution. Users enter $\log_2(\text{TPM} + 1)$ values for the eight genes and obtain a real-time logit $P(\text{DM2})$ score. Three preset profiles (TCGA median, DM1 prototype, DM2 prototype) allow non-computational reviewers to verify the calculator without input data. No server, no installation, no telemetry — intended both as a transparency artefact and as a usable clinical-translation tool.

External validation on GSE76039 ($n = 37$ PDTC + ATC) yields direction-correct AUC 0.974, confirming the panel's transferability. A composite hot/cold immune score (cytolytic activity + IFN- γ + immune fraction) cleanly separates DM1 (hot) from DM2 (cold) with **Cohen's $d = +1.683$ (Mann-Whitney $p = 3.99 \times 10^{-18}$)**.

Liu-Xing four-genotype integration with honest robustness audit. Beyond the DM1/DM2 axis, we recover 36 TERT-promoter-mutated patients via the cBioPortal `thca_tcga_pub` mirror (Sanger-validated by TCGA Cell 2014). A four-group stratification (BRAF / RAS / TERT⁺ / triple-negative) yields multivariate logrank $p = 3.78 \times 10^{-5}$ for overall survival; 1,000-iteration bootstrap median $p = 2.6 \times 10^{-5}$ and leave-one-out worst-case $p < 10^{-3}$ across all 36 TERT⁺ patients confirm the separation is not single-event-driven. We honestly report that the univariate TERT signal (Cox HR = 6.31, $p = 3 \times 10^{-4}$) drops to multivariate HR = 1.88 ($p = 0.29$) after stage + age + sex adjustment, and reframe TERT⁺ as a clinically actionable molecular handle for Stage III/IV identification rather than a stage-and-age-independent prognostic marker. Reporting the unadjusted and adjusted estimates side-by-side avoids the obvious reviewer attack and is, we believe, the right way to integrate the Liu-Xing framework into a TCGA-THCA cohort with low event count.

The manuscript matches *npj Precision Oncology*'s scope on three fronts. First, the headline contribution is a **directly clinically deployable biomarker** with quantified outperformance over the standard-of-care molecular test (BRAF V600E), now corroborated by a Decision Curve Analysis showing dominance across the entire clinical threshold range and by 4-cohort meta-analytic

transferability. Second, the underlying transcriptomic axis (DM1/DM2) is statistically distinct from BRAF/RAS mutation status (Spearman $\rho = 0.49$ with the canonical dichotomy; 17 of 335 mutation-carrying tumours violate the canonical map) and maps to a hot/cold immune landscape with implications for both immunotherapy and TROP2-directed antibody-drug conjugate (ADC) stratification. Third, we propose evaluation as a correlative biomarker overlay in two ongoing thyroid TROP2-ADC trials (NCT06235216 SETHY, NCT07521670 STRAP), both of which currently accrue without molecular sub-stratification — translating the in-silico work into an immediately actionable trial-design recommendation.

This work represents a fully in-silico discovery from an independent Korean computational researcher (with part-time PhD affiliation at Seoul National University Graduate School of Convergence Science and Technology), in collaboration with Yu Hyeong Won (Department of Surgery, Seoul National University Bundang Hospital, Seongnam, Republic of Korea). Korean cohort cross-validation via Seoul National University Hospital / Bundang Hospital is in active outreach for the revision round; the v4 4-cohort $I^2 = 0\%$ result strengthens but does not replace this independent prospective test.

We honestly report sixteen limitations: no wet-lab validation, no Korean cohort yet integrated, Cox HR for DM1/DM2 not significant for OS (consistent with TCGA-THCA's exceptional prognosis), 2/5 robust external transfer, scRNA cohort limited to 7 patients, PRISM cell-line drug screen underpowered at $n = 5$ vs $n = 5$, Thorsson cross-reference replaced by our composite, pan-cancer transfer is signature-overlap only, TERT calls from cBioPortal mirror (not BAM re-call), TERT subset only 6 events (with bootstrap and leave-one-out audit reported), the stage-adjustment caveat (univariate HR 6.31 $\rightarrow$ multivariate HR 1.88), the 4 meta-analysis cohorts overlap with axis discovery via GSE76039, DCA dominance is a population-level result with calibration as the next individual-decision step, **TCGA per-patient ^{131}I dose recorded for only 36/513 patients (UCSC Xena legacy clinical recovery; cBioPortal pan-can-atlas mirror does not retain this field), chr7 finding borderline at $\alpha = 0.05$ with bootstrap CI [0.000, 0.079], and MSK 2016 chr7 reference returns sparse GISTIC calls (targeted-panel artefact) so the chr7 finding is reported TCGA-only with the explicit external-replication-pending caveat. The 8-gene panel ΔAUC outperformance, now triply corroborated by meta-analysis + DCA + subgroup forest, is the central submission-defensible contribution.

Suggested reviewers (no conflicts of interest declared): - **Yuri Nikiforov, MD, PhD** — University of Pittsburgh — thyroid genomic classification. - **James A. Fagin, MD** — Memorial Sloan Kettering Cancer Center — BRAF biology and PTC molecular subtyping. - **Ricardo R. Lima, PhD** — PUC-RJ — thyroid sub-

classification and Latin American cohorts. - **Mingzhao Xing, MD, PhD** — Johns Hopkins / SUSTech — Liu-Xing four-genotype prognostic framework foundational to our R8 integration.

All raw outputs, pipeline scripts, and reproducibility archives are available in the submission package (anonymous_code.zip for double-blind review). The interactive calculator and BOOST-sprint validation tables are at reports/v17_boost/ and results/v17_boost/. The authors declare no competing interests. This work has not been submitted elsewhere.

Sincerely,

Co-corresponding authors:

Seungho Cook Independent Researcher (with part-time PhD affiliation), Seoul, Republic of Korea kukshomr@gmail.com

Yu Hyeong Won Department of Surgery, Seoul National University Bundang Hospital, Seongnam, Republic of Korea